

We'll Cross That Bridge When We Get There: Ancestral Character Reconstruction with Monte Carlo Diffusion Bridge Sampling

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Introduction

- ATR is the process of reconstruction historical trait values e.g tooth length over periods of deep time in reference to a phylogeny
- Requires a model of evolution
- Normally BM or OU
- Ancestral Trait reconstruction has a number of issues:
 - Normality
 - Markov
- We take advantage of SDE parametrization to extend these

$$X_t = b(X_t, t)dt + \sigma(X_t, t)dB_t$$

Methods

- Explain different SDE versions
- Bridge allows us to condition these SDEs
- We can therefore simulate large numbers of bridges conditioned on our leaves
- Perform a pruning Algorithm as seen in figure 2

Results

- We Performed our bridge algorithm with both traditional BM and OU models on a large marsupial data set leading to the tree seen in figure 1
- Achieves similar results to those seen with Phytools
- Drift and Diffusion function can be easily updated for new models
- Scaling to multivariate case is trivial
- Optimizing rate parameters remains a difficulty

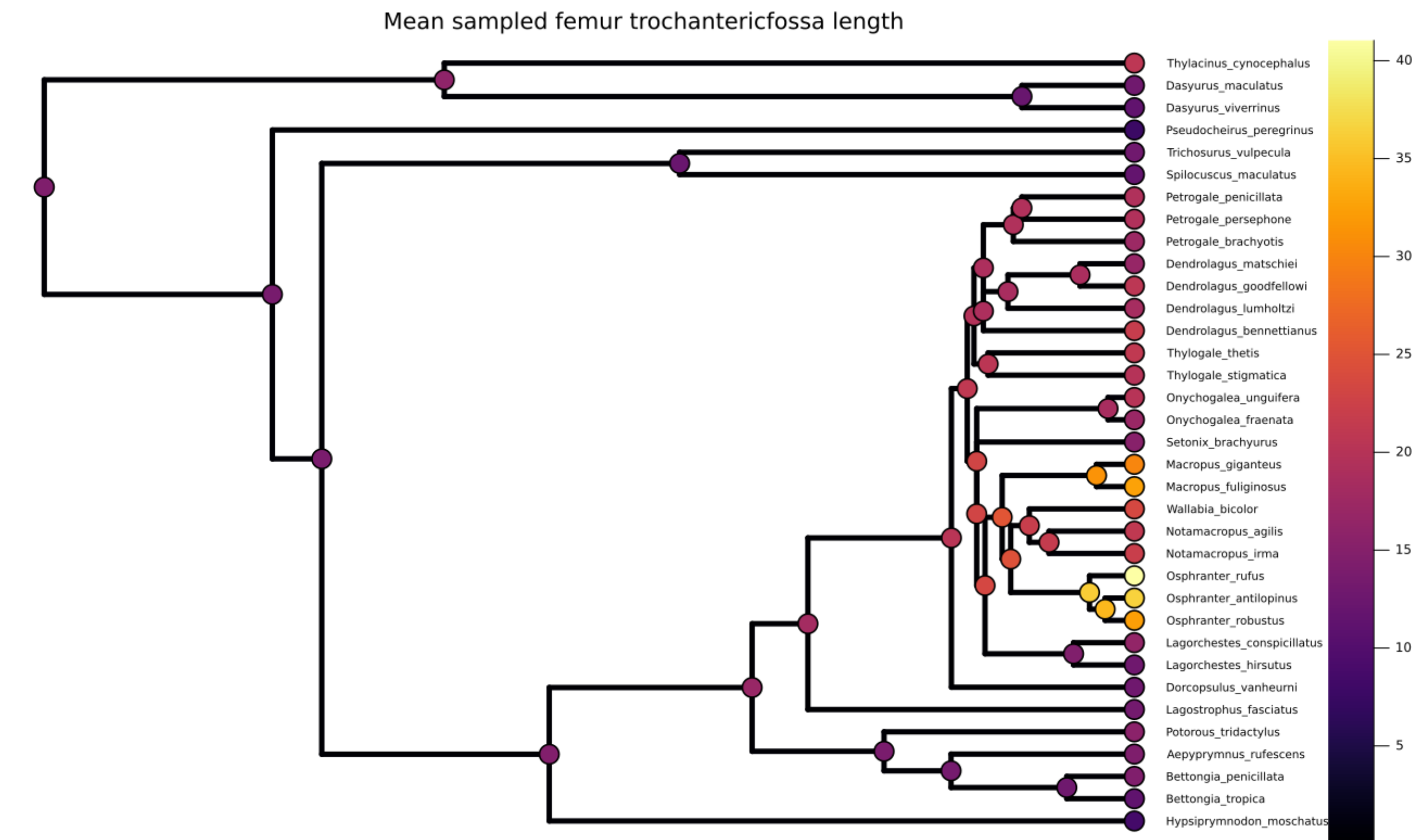
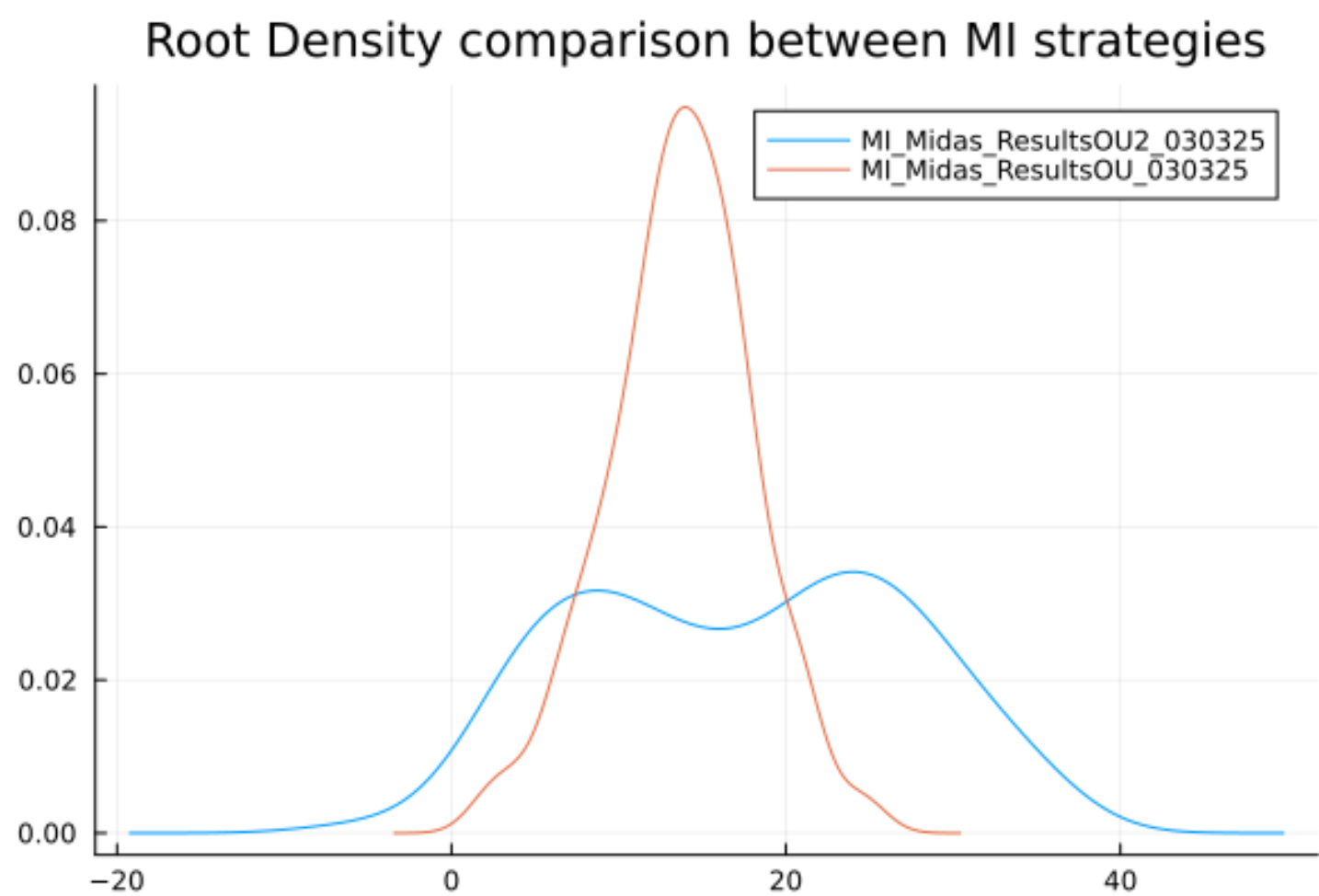
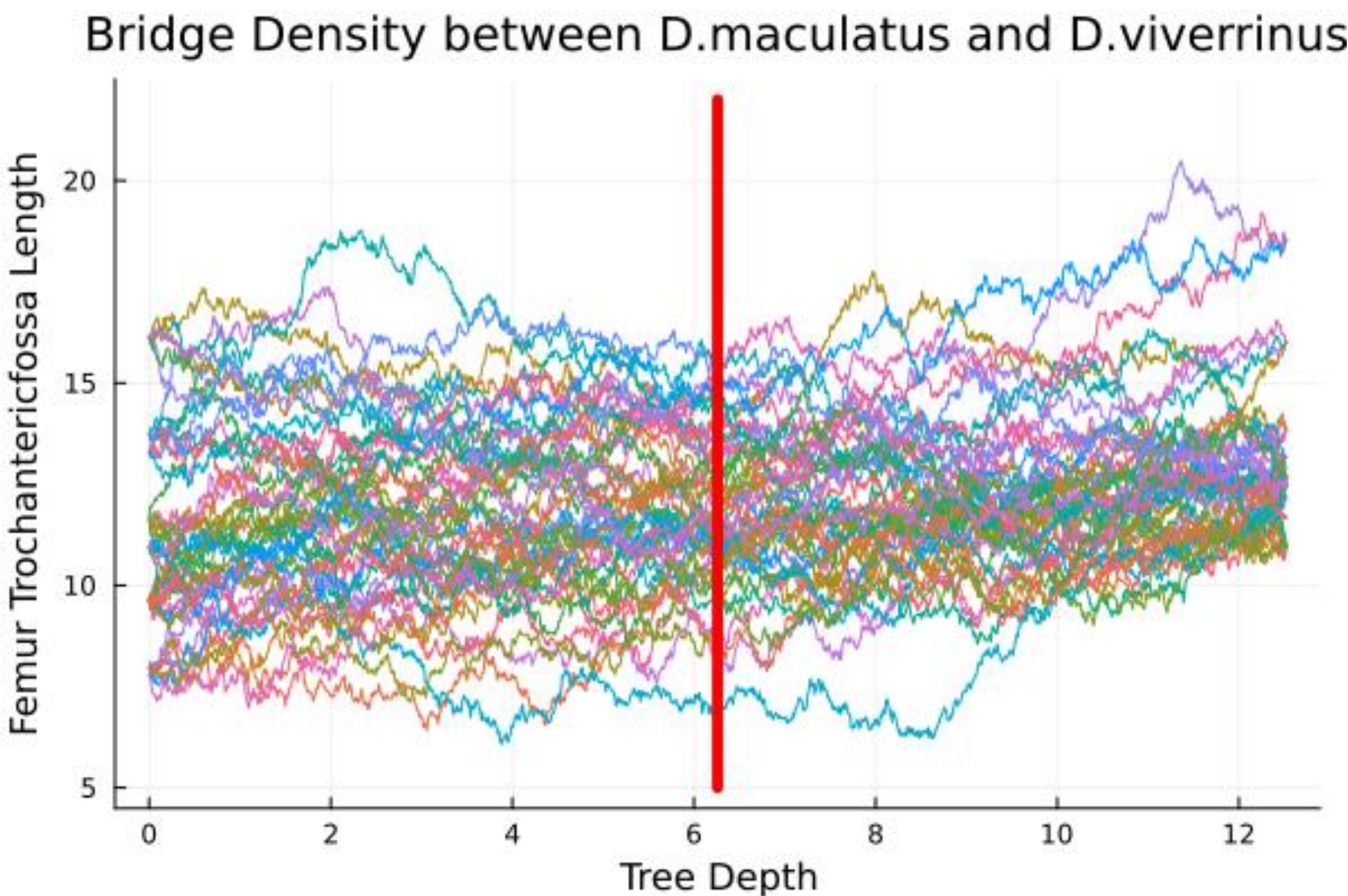


Figure 3: Ancestral trait reconstruction of 35 marsupial species for femur trochanteric fossa length value from 25 multiple imputations, from 25 independent initial samples with trait reconstruction repeated 25 times indicates the mean with heat map over the marker indicated predicted mean values



Parental
Density 1
+ Pictures



Parental
Density 2 +
Pictures

Figure 2: Brownian Bridge trait reconstruction between *Dasyurus maculatus* and *Dasyurus viverrinus*. **A)** contributing Brownian Bridges to the ancestral trait value of the ancestor, red line indicates the time point of the ancestor **B)** kernel density estimates of the contributing leaves to the resulting ancestral nodes, blue line corresponds to the density seen in figure A.